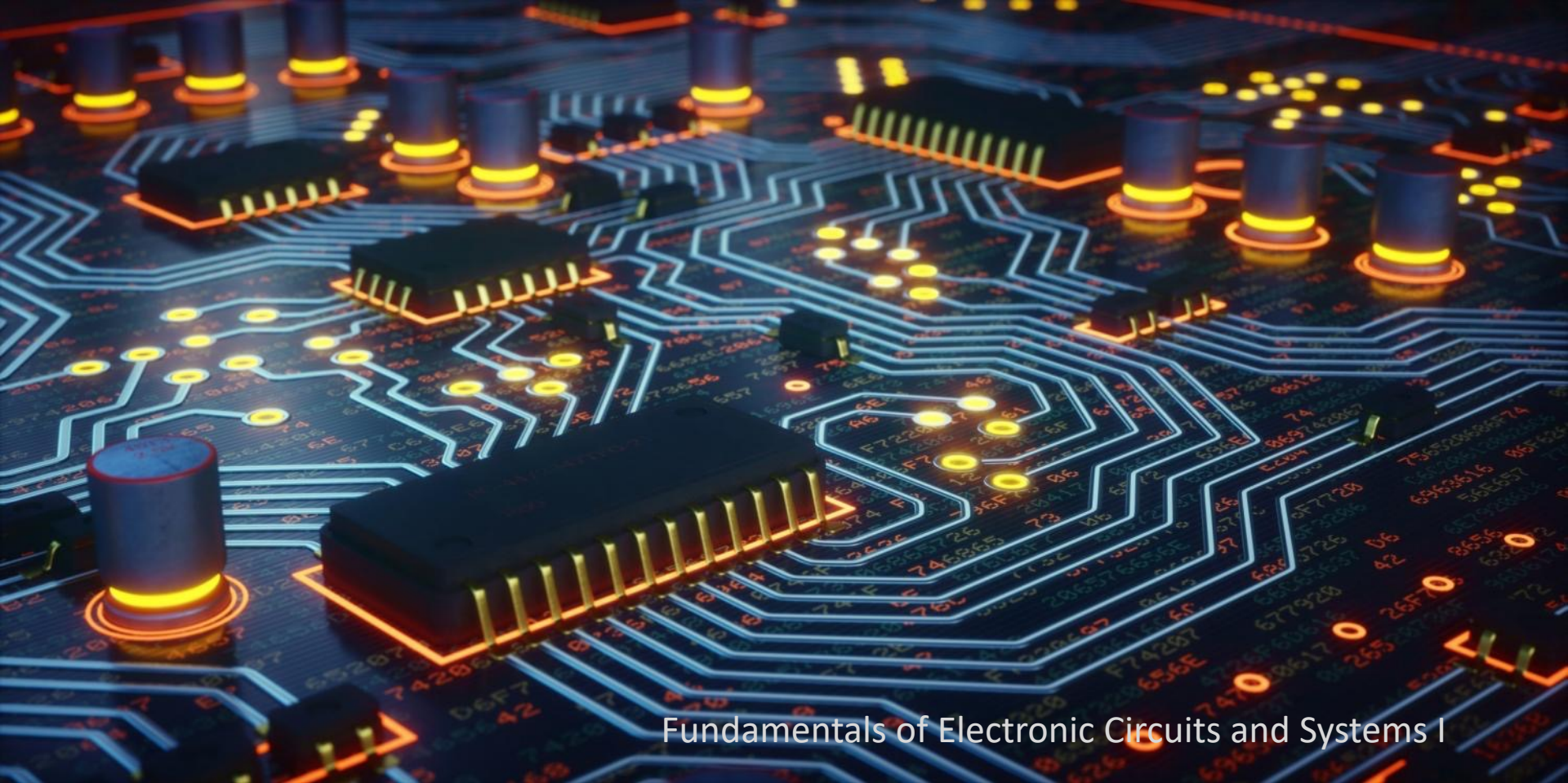




Fundamentals of Electronic Circuits and Systems I

Transient Circuit Analysis

Milin Zhang

Dept of EE, Tsinghua University



Outlines

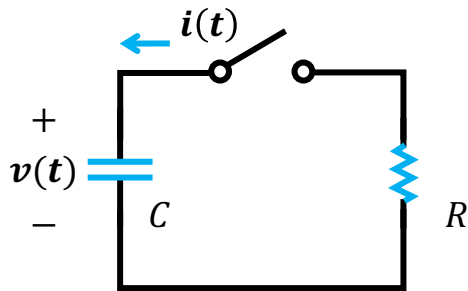
- 1st order circuit
- 2nd order circuit

Recall: Capacitor & Inductor

			
i - v characteristic	$i = \frac{v}{R}$	$i(t) = C \frac{dv(t)}{dt}$	$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v(x) dx$
v - i characteristic	$v = iR$	$v(t) = v(t_0) + \frac{1}{C} \int_{t_0}^t i(x) dx$	$v(t) = L \frac{di(t)}{dt}$
p (power transferred in)	$p = vi$	$p = vi$	$p = vi$
w (stored energy)	0	$w = \frac{1}{2} C v^2(t) = \frac{Q^2(t)}{2C}$	$w = \frac{1}{2} L i^2(t) = \frac{\lambda^2(t)}{2L}$
Series combination	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel combination	$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$
DC behavior	NO	open circuit	short circuit
Instantaneous change of v	$\checkmark$	$\times$	$\checkmark$
Instantaneous change of i	$\checkmark$	$\checkmark$	$\times$

Example 1: Source free RC circuit

QUESTION: Assume the capacitor C has been charged to V_0 before the switch is turned on. Find the response after the switch is turned on.



- Assume the switch is turned on @ $t = 0$
- According to KCL

$$i_C = i_R$$

$$\rightarrow C \frac{d}{dt} v(t) + \frac{v(t)}{R} = 0$$

$$\rightarrow \frac{d}{dt} v(t) + \frac{v(t)}{RC} = 0$$

1st order differential equation

Recall: Calculus

QUESTION: Find the solution $x(t)$ of the following equation

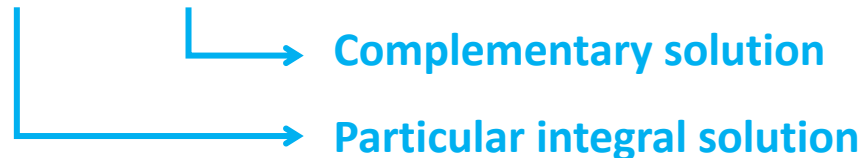
$$\frac{d}{dt}x(t) + a_1x(t) = f(t) \quad \leftarrow \text{General form of 1st order differential equation}$$

- Step 1: Find any solution to the original equation $x(t) = x_p(t)$
- Step 2: Find a solution $x(t) = x_c(t)$ to the homogeneous equation

$$\frac{d}{dt}x(t) + a_1x(t) = 0$$

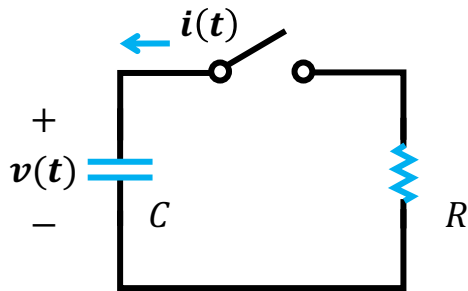
- Step 3: the solution to the original equation can be written as

$$x(t) = x_p(t) + x_c(t)$$



Example 1: Source free RC circuit

QUESTION: Assume the capacitor C has been charged to V_0 before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}v(t) + \frac{v(t)}{RC} = 0$$

- Step 1a: find the **particular integral solution** $v_p(t)$

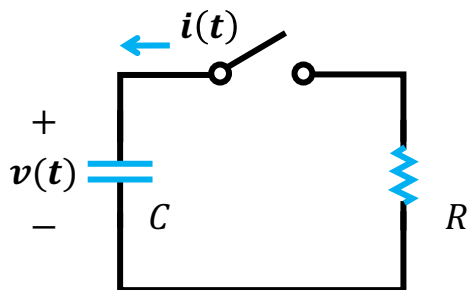
$$\text{Assume } v_p(t) = K_1$$

- Step 1b: substitute $i_p(t)$ to the equation

$$\frac{K_1}{RC} = 0 \quad \rightarrow \quad K_1 = 0$$

Example 1: Source free RC circuit

QUESTION: Assume the capacitor C has been charged to V_0 before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}v(t) + \frac{v(t)}{RC} = 0$$

- Step 2a: find the **homogeneous equation**

$$\frac{d}{dt}v(t) + a_1v(t) = 0 \quad \text{where } a_1 = \frac{1}{RC}$$

- Step 2b: find the **complementary solution** $v_c(t)$ to the homogeneous equation

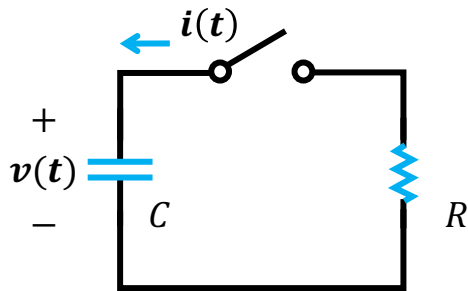
$$v_c(t) = K_2 e^{-st}$$

Particular integral solution

$$v_p(t) = 0$$

Example 1: Source free RC circuit

QUESTION: Assume the capacitor C has been charged to V_0 before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}v(t) + \frac{v(t)}{RC} = 0$$

- Step 2c: substitute $v_c(t)$ to the homogeneous equation

$$\frac{d}{dt}(K_2 e^{-st}) + a_1(K_2 e^{-st}) = 0$$

$$\rightarrow -sK_2 e^{-st} + a_1 K_2 e^{-st} = 0$$

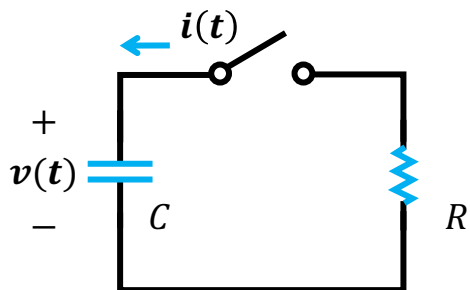
$$\rightarrow s = a_1 \quad \text{where } a_1 = \frac{1}{RC}$$

Particular integral solution

$$v_p(t) = 0$$

Example 1: Source free RC circuit

QUESTION: Assume the capacitor C has been charged to V_0 before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}v(t) + \frac{v(t)}{RC} = 0$$

- Step 3a: find the value of $v(t)$ at one instant of time

$$v(0) = V_0$$

- Step 3b: substitute $v(0) = V_0$ to $v(t)$

$$v(t) = K_2 e^{-a_1 t}$$

$$\rightarrow K_2 = V_0$$

Particular integral solution

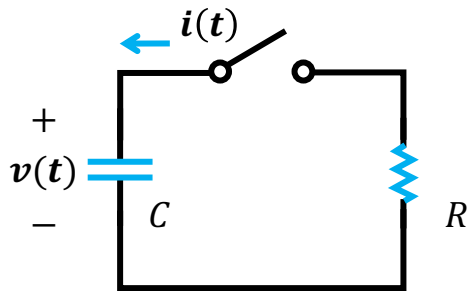
$$v_p(t) = 0$$

Complementary solution

$$v_c(t) = K_2 e^{-a_1 t}$$

Example 1: Source free RC circuit

QUESTION: Assume the capacitor C has been charged to V_0 before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}v(t) + \frac{v(t)}{RC} = 0$$

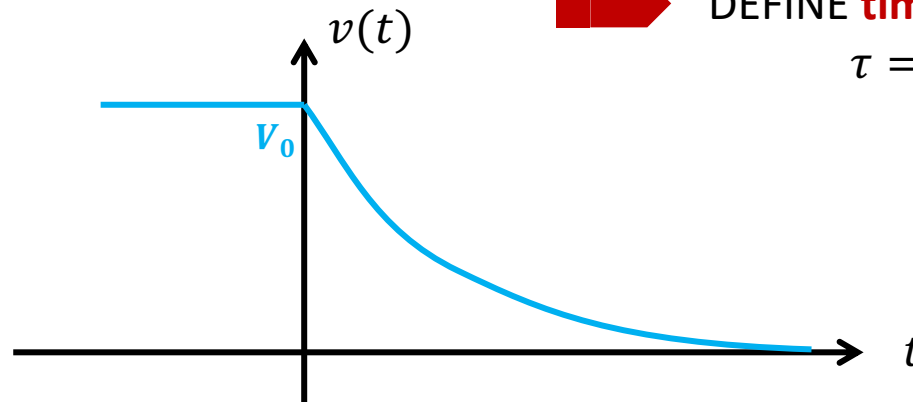
Solution: $v(t) = V_0 e^{-\frac{1}{RC}t}$

"—" indicates that the voltage across the capacitor is decaying exponentially



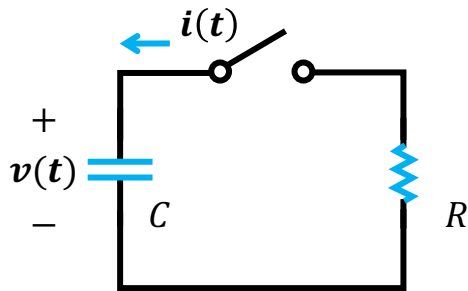
DEFINE **time constant**
 $\tau = RC$

The circuit reaches steady-state faster, if a smaller time constant is applied



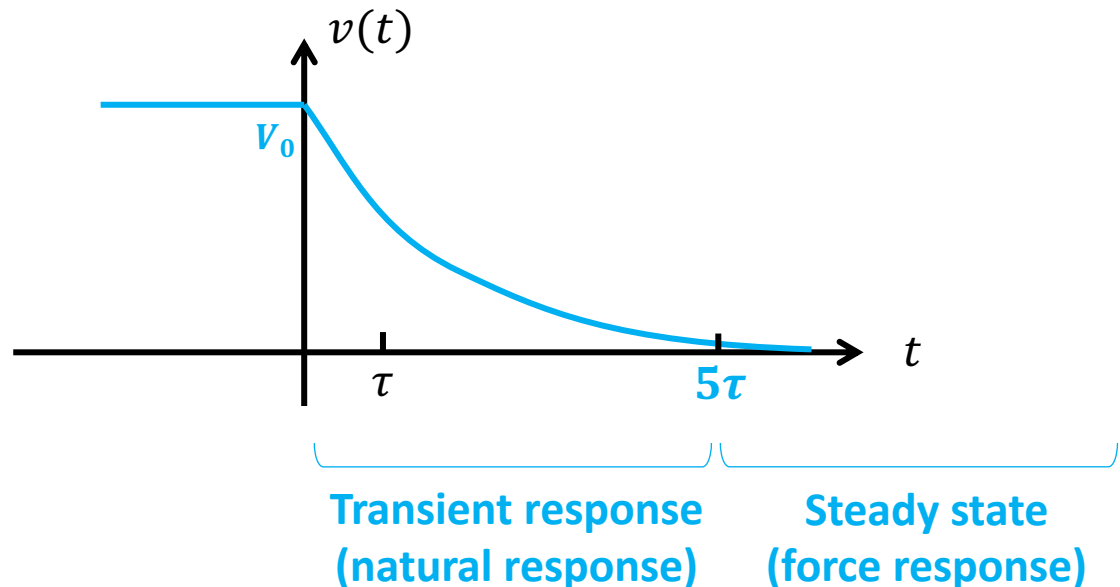
Example 1: Source free RC circuit

QUESTION: Assume the capacitor C has been charged to V_0 before the switch is turned on. Find the response after the switch is turned on.



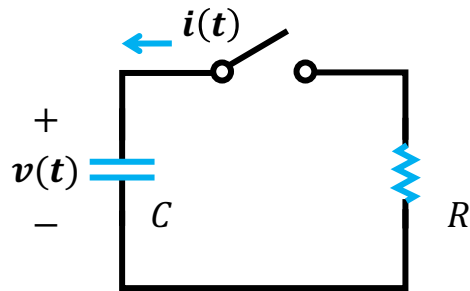
- When $t = \tau$, $v(t) = 0.368V_0$
- When $t = 5\tau$, $v(t) = 0.0067V_0$

$$\left[\begin{array}{l} \frac{d}{dt}v(t) + \frac{v(t)}{RC} = 0 \\ v(t) = V_0 e^{-\frac{1}{\tau}t} \\ \tau = RC \end{array} \right]$$

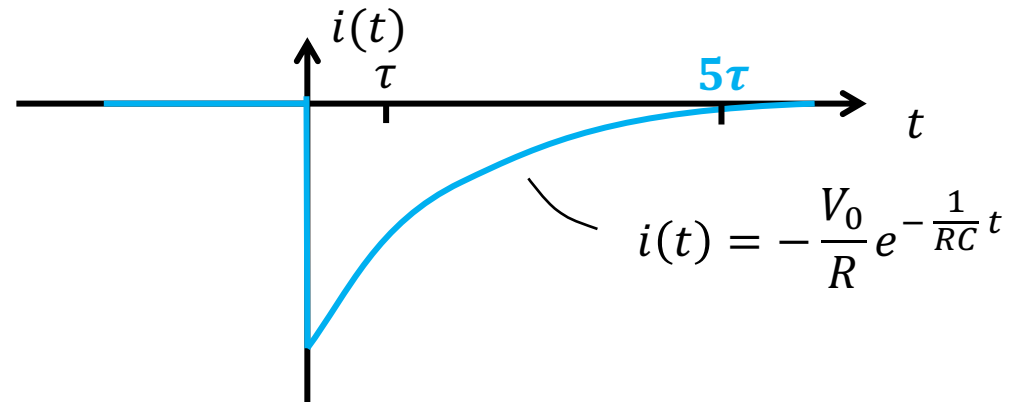
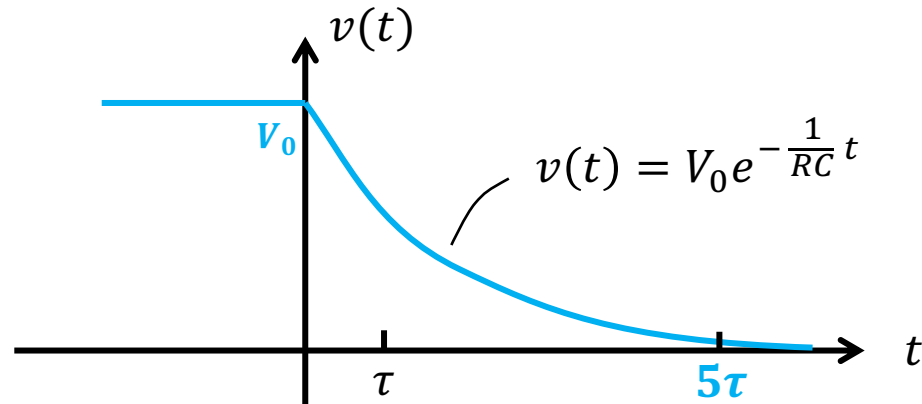


Example 1: Source free RC circuit

QUESTION: Assume the capacitor C has been charged to V_0 before the switch is turned on. Find the response after the switch is turned on.

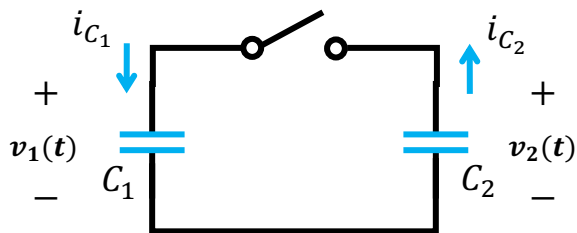


Voltage on capacitor
CANNOT change abruptly,
but current can.



Recall: an example from Chapter 4

QUESTION: Find the total energy stored in the two capacitors before & after the switch is turned on, if the charges on the two capacitors at $t = 0$ are $q_1(0) = Q_1$ and $q_2(0) = Q_2$, respectively.



- BEFORE the switch is turned on ($t < 0$)

$$w(t < 0) = \frac{Q_1^2}{2C_1} + \frac{Q_2^2}{2C_2}$$

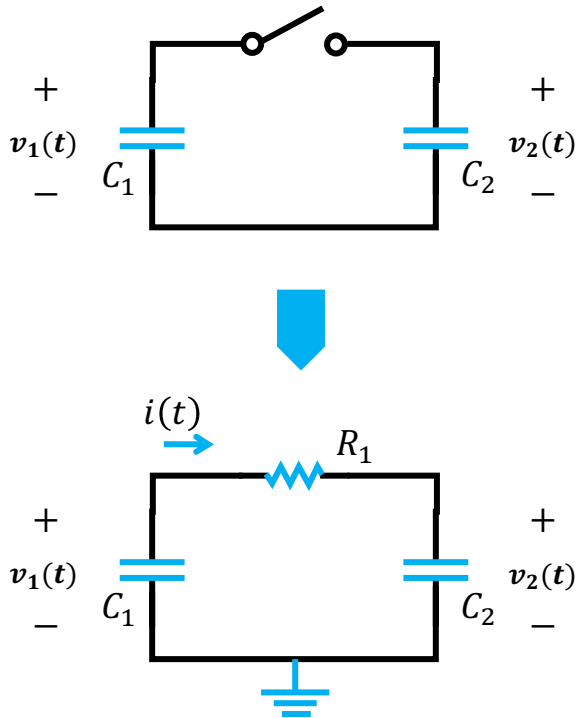
- AFTER the switch is turned on ($t \geq 0$). Assume the circuit is in steady state @ $t = t_1$

$$w(t_1) = \frac{(Q_1 + Q_2)^2}{2(C_1 + C_2)}$$

WHY $w(t < 0) \neq w(t_1)$?

Example 2

QUESTION: Find the total energy stored in the two capacitors before & after the switch is turned on, if the charges on the two capacitors at $t = 0$ are $q_1(0) = Q_1$ and $q_2(0) = Q_2$, respectively. The resistance of the switch is R_1 when it is turned on.



- After the switch is turned on, according to KVL

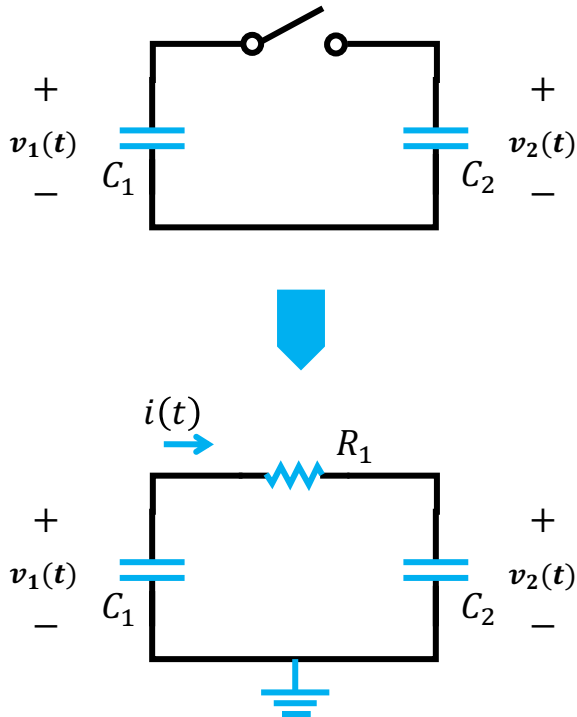
$$\frac{1}{C_1} \int_{-\infty}^t i(t) dt + i(t)R_1 + \frac{1}{C_2} \int_{-\infty}^t i(t) dt = 0$$

$$\rightarrow \frac{d}{dt} i(t) + \frac{1}{R_1} \left(\frac{1}{C_1} + \frac{1}{C_2} \right) i(t) = 0$$

$$\text{Define } a_1 = \frac{1}{R_1} \left(\frac{1}{C_1} + \frac{1}{C_2} \right)$$

Example 2

QUESTION: Find the total energy stored in the two capacitors before & after the switch is turned on, if the charges on the two capacitors at $t = 0$ are $q_1(0) = Q_1$ and $q_2(0) = Q_2$, respectively. The resistance of the switch is R_1 when it is turned on.



$$\frac{d}{dt}i(t) + a_1 i(t) = 0$$

- Step 1: find the particular integral solution

$$i_p(t) = 0$$

- Step 2: find the complementary solution

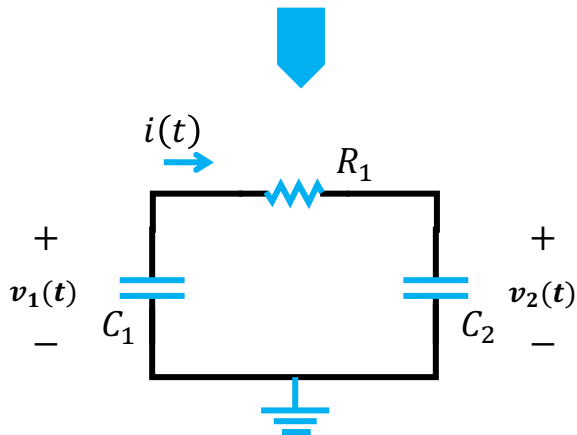
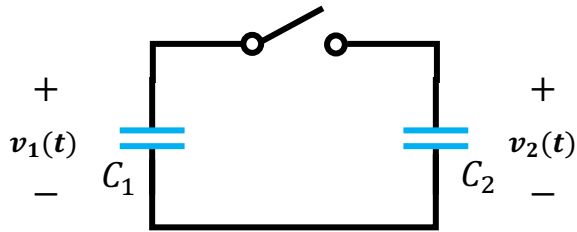
$$i_c(t) = K_2 e^{-a_1 t}$$

- Step 3: find the value of $i(t)$ at one instant of time

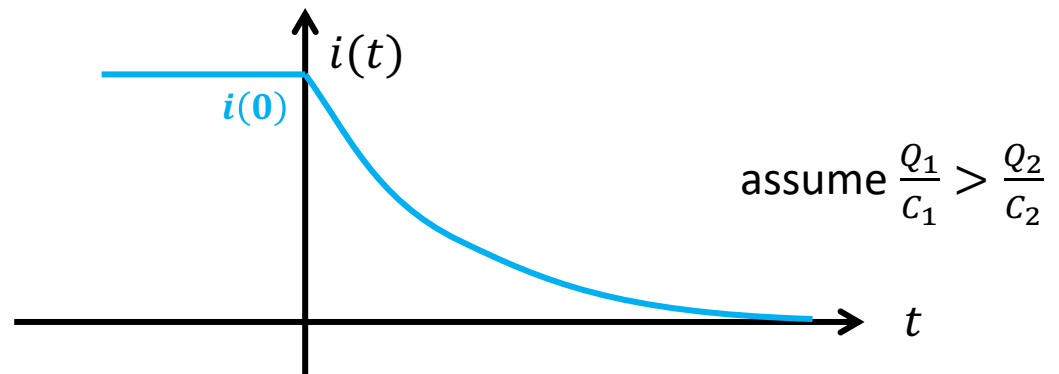
$$i(0) = \frac{1}{R_1} \left(\frac{Q_1}{C_1} - \frac{Q_2}{C_2} \right) \rightarrow K_2 = i(0)$$

Example 2

QUESTION: Find the total energy stored in the two capacitors before & after the switch is turned on, if the charges on the two capacitors at $t = 0$ are $q_1(0) = Q_1$ and $q_2(0) = Q_2$, respectively. The resistance of the switch is R_1 when it is turned on.



$$i(t) = i(0)e^{-a_1 t} \quad \text{where} \quad \begin{cases} a_1 = \frac{1}{R_1} \left(\frac{1}{C_1} + \frac{1}{C_2} \right) \\ i(0) = \frac{1}{R_1} \left(\frac{Q_1}{C_1} - \frac{Q_2}{C_2} \right) \end{cases}$$

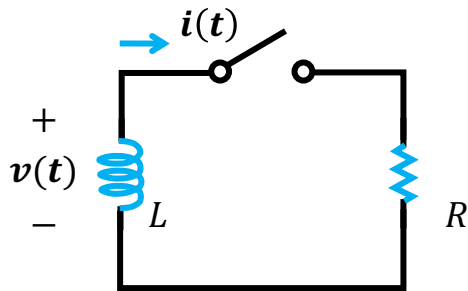


Outlines

- 1st order circuit
 - Source free RC circuit
 - **Source free RL circuit**

Example 3: Source free RL circuit

QUESTION: Assume the capacitor L has been charged to I_0 before the switch is turned on. Find the response after the switch is turned on.



- Assume the switch is turned on @ $t = 0$
- According to KVL

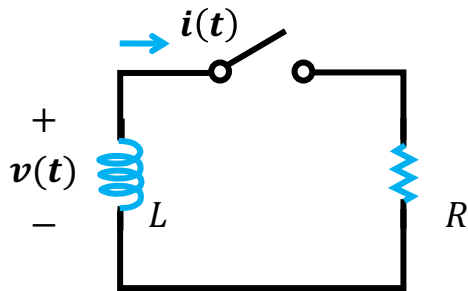
$$L \frac{d}{dt} i(t) + i(t)R = 0$$

$$\rightarrow \frac{d}{dt} i(t) + \frac{R}{L} i(t) = 0$$

1st order differential equation

Example 3: Source free RL circuit

QUESTION: Assume the capacitor L has been charged to I_0 before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = 0$$

- Step 1a: find the **particular integral solution** $i_p(t)$

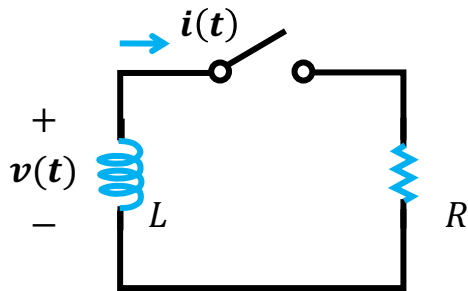
$$\text{Assume } i_p(t) = K_1$$

- Step 1b: substitute $i_p(t)$ to the equation

$$\frac{RK_1}{L} = 0 \quad \rightarrow \quad K_1 = 0$$

Example 3: Source free RL circuit

QUESTION: Assume the capacitor L has been charged to I_0 before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = 0$$

- Step 2a: find the **homogeneous equation**

$$\frac{d}{dt}i(t) + a_1 i(t) = 0 \quad \text{where } a_1 = \frac{R}{L}$$

- Step 2b: find the **complementary solution** $i_c(t)$ to the homogeneous equation

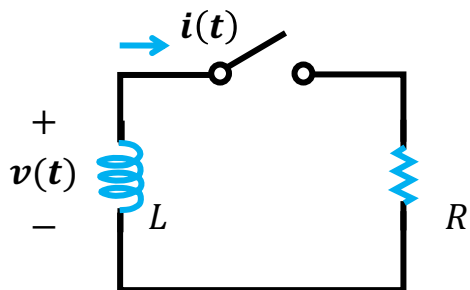
$$v_c(t) = K_2 e^{-st}$$

Particular integral solution

$$i_p(t) = 0$$

Example 3: Source free RL circuit

QUESTION: Assume the capacitor L has been charged to I_0 before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = 0$$

- Step 2c: substitute $v_c(t)$ to the homogeneous equation

$$\frac{d}{dt}(K_2 e^{-st}) + a_1(K_2 e^{-st}) = 0$$

$$\rightarrow -sK_2 e^{-st} + a_1 K_2 e^{-st} = 0$$

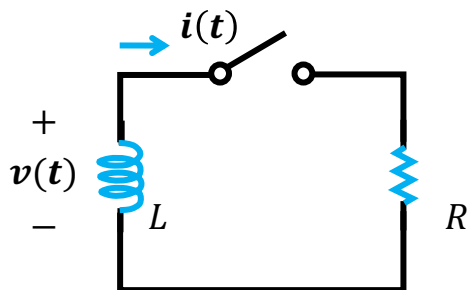
$$\rightarrow s = a_1 \quad \text{where } a_1 = \frac{R}{L}$$

Particular integral solution

$$i_p(t) = 0$$

Example 3: Source free RL circuit

QUESTION: Assume the capacitor L has been charged to I_0 before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = 0$$

- Step 3a: find the value of $i(t)$ at one instant of time

$$i(0) = I_0$$

- Step 3b: substitute $i(0) = I_0$ to $i(t)$

$$i(t) = K_2 e^{-a_1 t}$$

$$\rightarrow K_2 = I_0$$

Particular integral solution

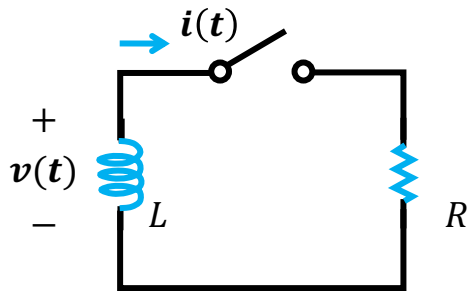
$$i_p(t) = 0$$

Complementary solution

$$i_c(t) = K_2 e^{-a_1 t}$$

Example 3: Source free RL circuit

QUESTION: Assume the capacitor L has been charged to I_0 before the switch is turned on. Find the response after the switch is turned on.

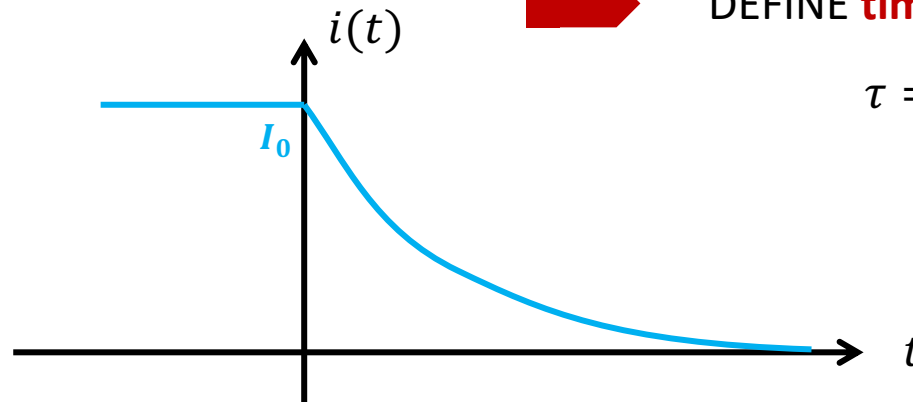


$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = 0$$

Solution: $i(t) = I_0 e^{-\frac{R}{L}t}$

"—" indicates that the current through the capacitor is decaying exponentially

The circuit reaches steady-state faster, if a smaller time constant is applied

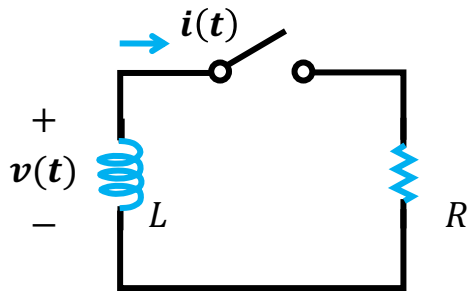


DEFINE **time constant**

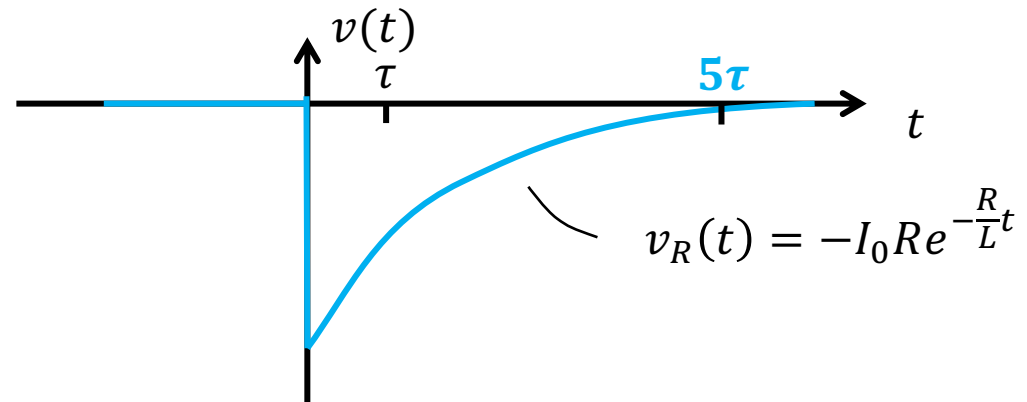
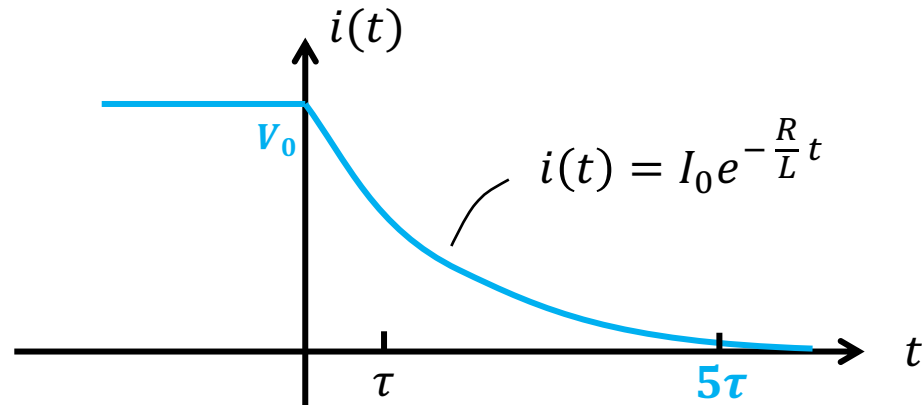
$$\tau = \frac{L}{R}$$

Example 3: Source free RL circuit

QUESTION: Assume the capacitor L has been charged to I_0 before the switch is turned on. Find the response after the switch is turned on.



Current through inductor
CANNOT change abruptly,
but voltage can.

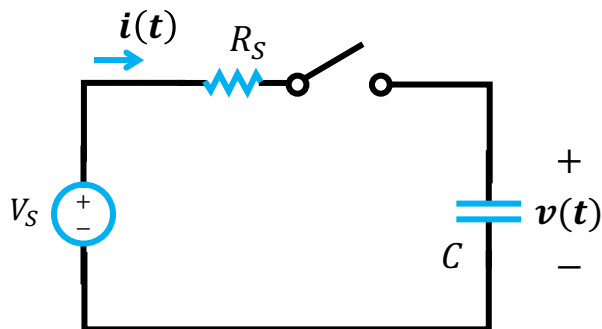


Outlines

- 1st order circuit
 - Source free RC circuit
 - Source free RL circuit
 - **Pulse response**

Example 4: charging of a capacitor

QUESTION: Assume there is no charge on the capacitor C before the switch is turned on. Find the response after the switch is turned on.



- Assume the switch is turned on @ $t = 0$
- According to KCL

$$i_C = i_R = i(t) = C \frac{d}{dt} v(t)$$

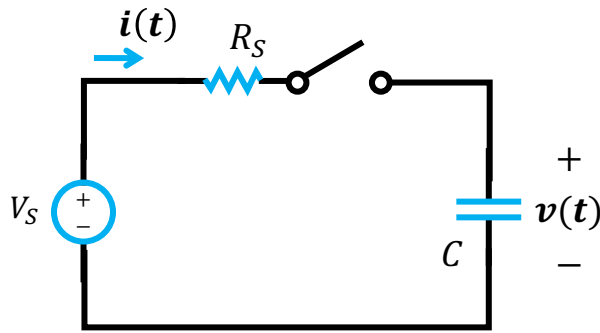
- According to KVL

$$i_R R + v(t) = V_S$$

$$\rightarrow RC \frac{d}{dt} v(t) + v(t) = V_S$$

Example 4: charging of a capacitor

QUESTION: Assume there is no charge on the capacitor C before the switch is turned on. Find the response after the switch is turned on.



$$\frac{d}{dt}v(t) + \frac{1}{RC}v(t) = \frac{1}{RC}V_s$$

- Step 1a: find the **particular integral solution $x_p(t)$**

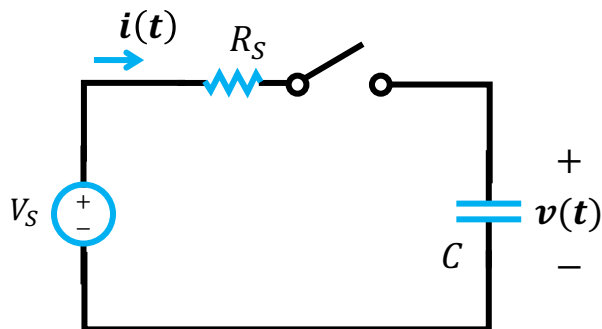
$$\text{Assume } x_p(t) = K_1$$

- Step 1b: substitute $x_p(t)$ to the equation

$$\frac{K_1}{RC} = \frac{1}{RC}V_s \quad \rightarrow \quad K_1 = V_s$$

Example 4: charging of a capacitor

QUESTION: Assume there is no charge on the capacitor C before the switch is turned on. Find the response after the switch is turned on.



Particular integral solution

$$v_p(t) = V_S$$

$$\frac{d}{dt}v(t) + \frac{1}{RC}v(t) = \frac{1}{RC}V_S$$

- Step 2a: find the **homogeneous equation**

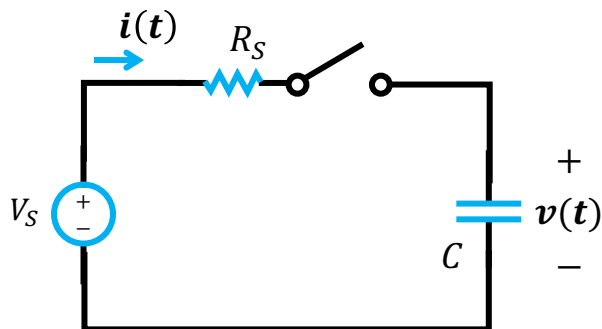
$$\frac{d}{dt}v(t) + a_1v(t) = 0 \quad \text{where } a_1 = \frac{1}{RC}$$

- Step 2b: find the **complementary solution** $v_c(t)$ to the homogeneous equation

$$v_c(t) = K_2e^{-a_1t}$$

Example 4: charging of a capacitor

QUESTION: Assume there is no charge on the capacitor C before the switch is turned on. Find the response after the switch is turned on.



Particular integral solution

$$v_p(t) = V_S$$

Complementary solution

$$v_c(t) = K_2 e^{-a_1 t}$$

Full solution $v(t) = v_p(t) + v_c(t)$

$$\frac{d}{dt} v(t) + \frac{1}{RC} v(t) = \frac{1}{RC} V_S$$

- Step 3a: find the value of $v(t)$ at one instant of time

$$v(0) = 0$$

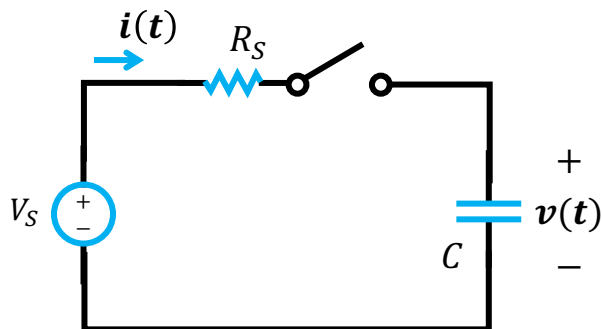
- Step 3b: substitute $v(0) = 0$ to $v(t)$

$$v(t) = V_S + K_2 e^{-a_1 t}$$

$$\rightarrow K_2 = -V_S$$

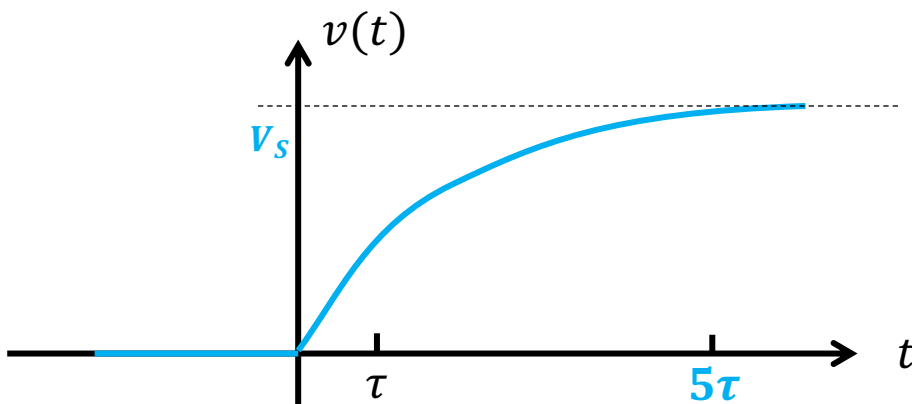
Example 4: charging of a capacitor

QUESTION: Assume there is no charge on the capacitor C before the switch is turned on. Find the response after the switch is turned on.



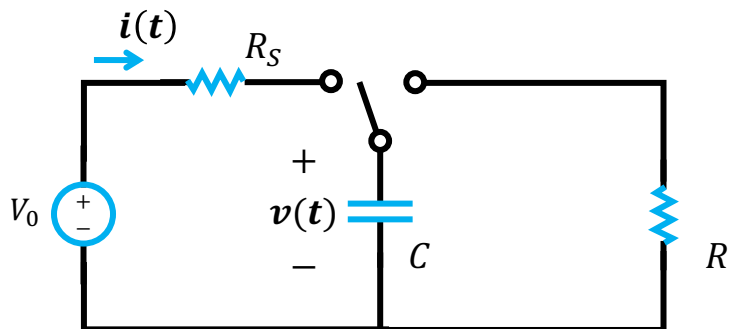
$$\frac{d}{dt}v(t) + \frac{1}{RC}v(t) = \frac{1}{RC}V_S$$

Solution: $v(t) = V_S - V_S e^{-\frac{1}{RC}t}$



Charging & discharging of a cap

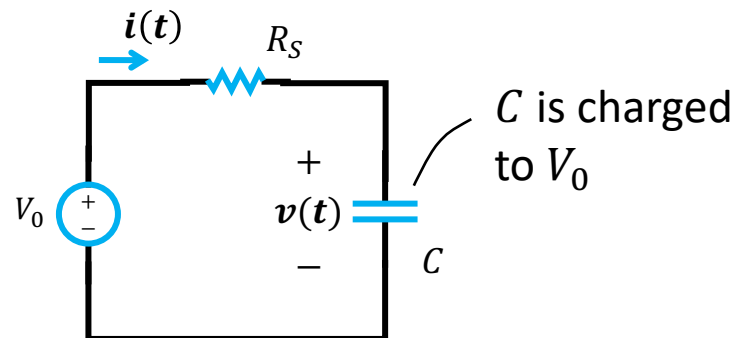
A more practical circuit



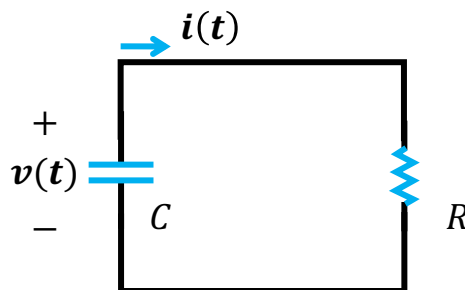
Key points:

- Find the initial voltage of the capacitor in phase 1
- Write the equations according to KCL/KVL
- Solve the differential equation

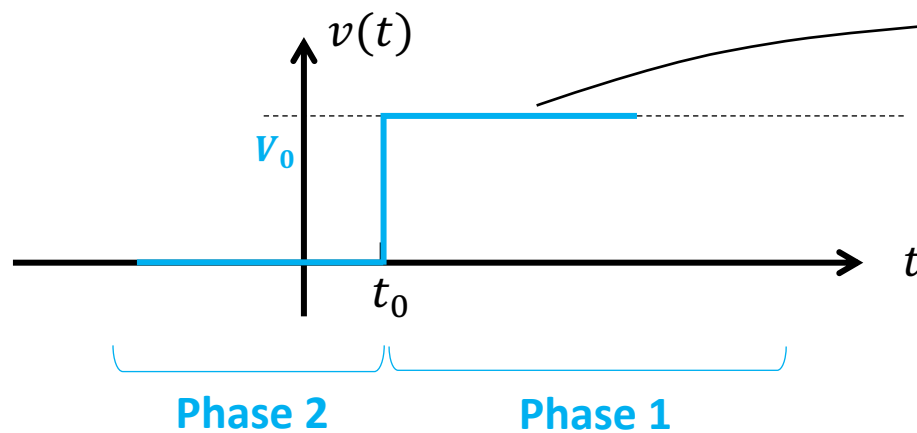
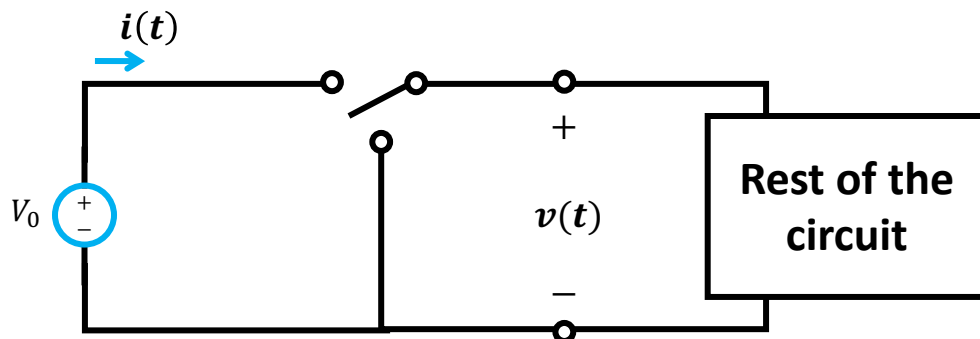
PHASE 1



PHASE 2



Step forcing function

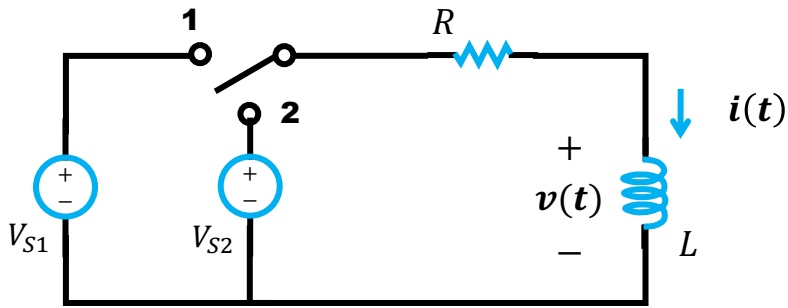


$$v(t) = V_0 u(t - t_0)$$

$$\text{where } u(t - t_0) = \begin{cases} 0 & t < t_0 \\ 1 & t \geq t_0 \end{cases}$$

Example 5: charging of an inductor

QUESTION: Assume there is no charge on the inductor L before the switch is turned on. Find the response after the switch is switching between node 1 and 2.

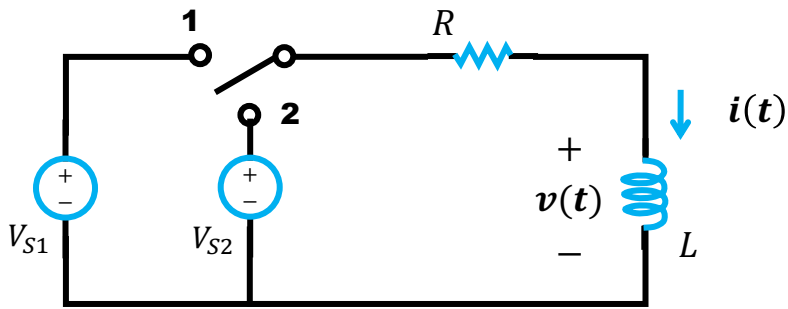


- When the switch turned to node 1, according to KVL

$$V_{S1} = i(t)R + L \frac{di(t)}{dt} \quad \rightarrow \quad \frac{di(t)}{dt} + i(t) \frac{R}{L} = \frac{V_{S1}}{L}$$

Example 5: charging of an inductor

QUESTION: Assume there is no charge on the inductor L before the switch is turned on. Find the response after the switch is switching between node 1 and 2.



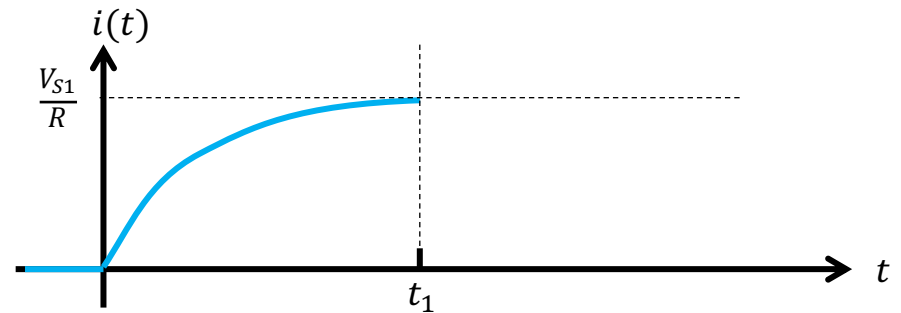
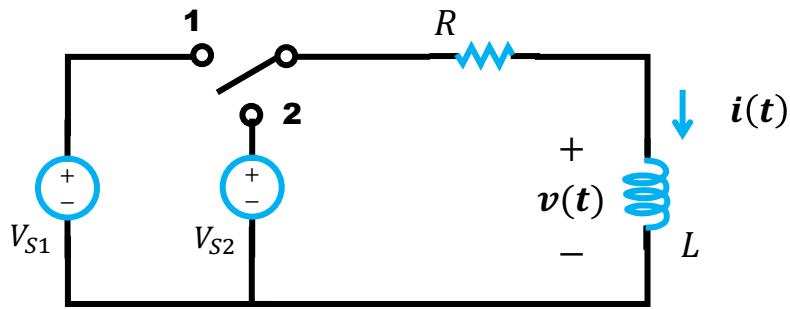
$$\frac{di(t)}{dt} + i(t) \frac{R}{L} = \frac{V_{S1}}{L}$$

- Step 1a: Assume the particular integral solution $i_p(t) = K_1$
- Step 1b: Substitute $i_p(t)$ to the equation $\rightarrow K_1 = \frac{V_{S1}}{R}$
- Step 2: Find the solution to the homogeneous equation

$$\frac{di(t)}{dt} + i(t) \frac{R}{L} = 0 \quad \text{Assume } i_c(t) = K_2 e^{-a_1 t} \quad \rightarrow \quad a_1 = \frac{R}{L}$$

Example 5: charging of an inductor

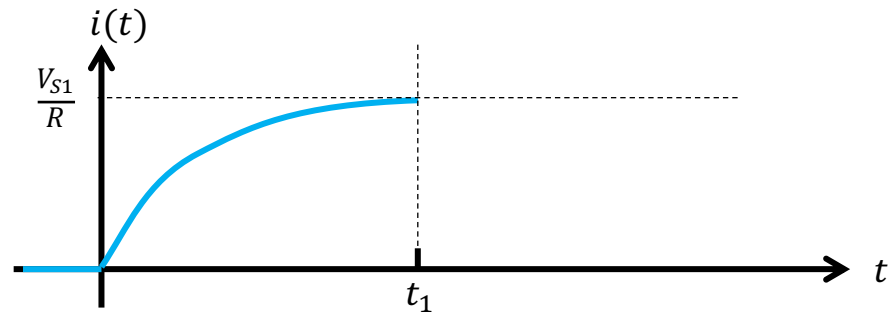
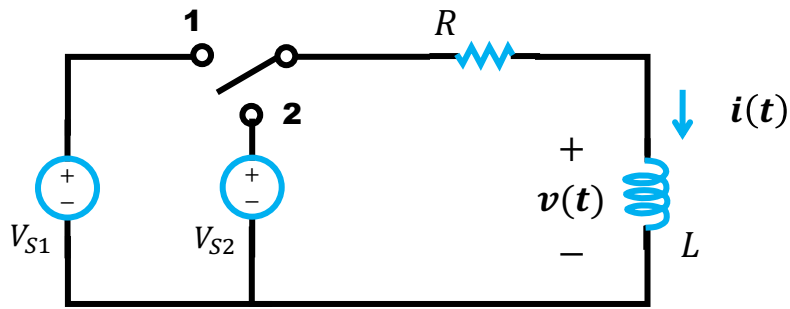
QUESTION: Assume there is no charge on the inductor L before the switch is turned on. Find the response after the switch is switching between node 1 and 2.



- The full solution $i(t) = i_p(t) + i_c(t) = \frac{V_{S1}}{R} + K_2 e^{-\frac{R}{L}t}$
- Step 3: find the value at one instant of time $i(0) = 0 \rightarrow K_2 = -\frac{V_{S1}}{R}$
- The full solution for $t \in [0, t_1)$ $i(t) = \frac{V_{S1}}{R} - \frac{V_{S1}}{R} e^{-\frac{R}{L}t}$

Example 5: charging of an inductor

QUESTION: Assume there is no charge on the inductor L before the switch is turned on. Find the response after the switch is switching between node 1 and 2.



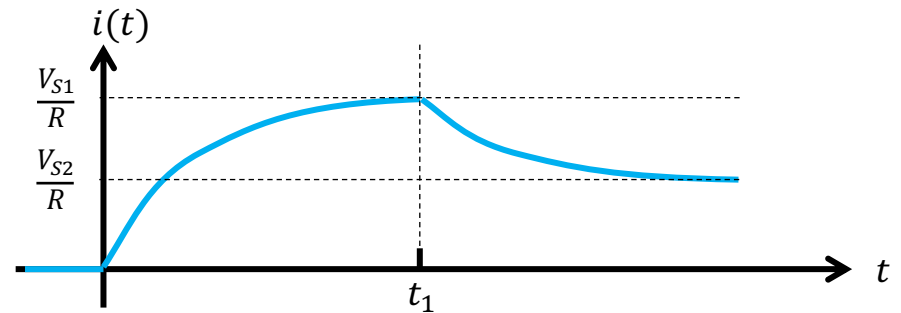
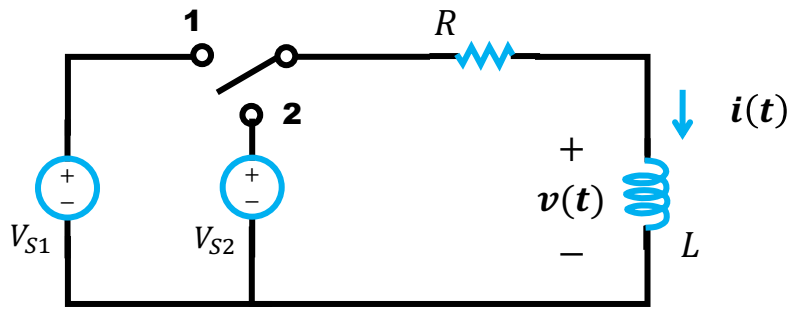
- Assume the switch turns to node 2 @ $t = t_1$
- According to KVL

$$V_{S2} = i(t)R + L \frac{di(t)}{dt} \rightarrow \frac{di(t)}{dt} + i(t) \frac{R}{L} = \frac{V_{S2}}{L}$$

- The full solution must be $i(t) = \frac{V_{S2}}{R} + K_2 e^{-\frac{R}{L}(t-t_1)}$

Example 5: charging of an inductor

QUESTION: Assume there is no charge on the inductor L before the switch is turned on. Find the response after the switch is switching between node 1 and 2.

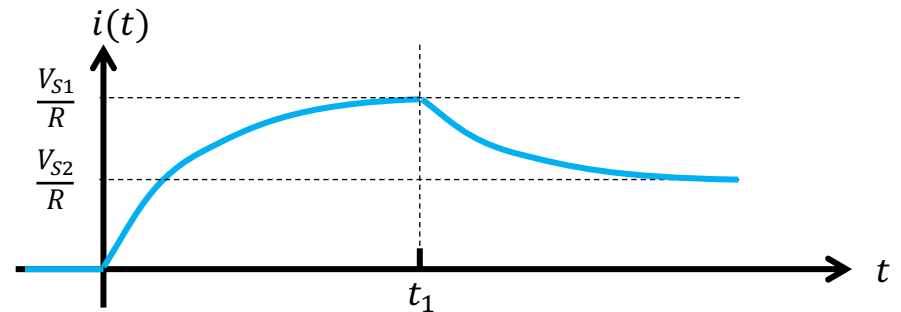
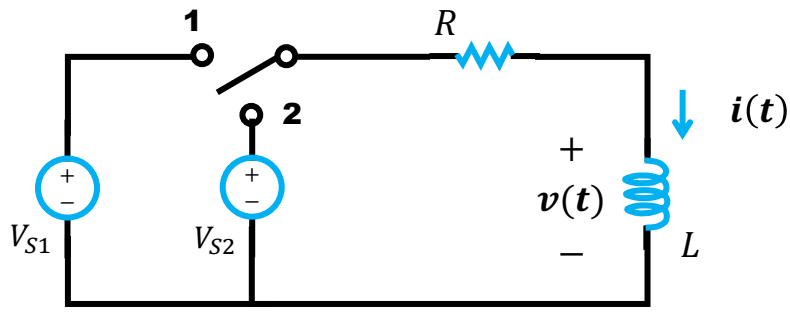


- find the value at one instant of time $i(t_1) = \frac{V_{S1}}{R} \rightarrow K_2 = \frac{V_{S1}}{R} - \frac{V_{S2}}{R}$
- The full solution for $t \geq t_1$

$$i(t) = \frac{V_{S2}}{R} + \left(\frac{V_{S1}}{R} - \frac{V_{S2}}{R} \right) e^{-\frac{R}{L}(t-t_1)}$$

Example 5: charging of an inductor

QUESTION: Assume there is no charge on the inductor L before the switch is turned on. Find the response after the switch is switching between node 1 and 2.



$$i(t) = \underbrace{\frac{V_{S2}}{R}}_{i(\infty)} + \underbrace{\left(\frac{V_{S1}}{R} - \frac{V_{S2}}{R} \right)}_{i(t_1) - i(\infty)} e^{-\frac{R}{L}(t-t_1)}$$

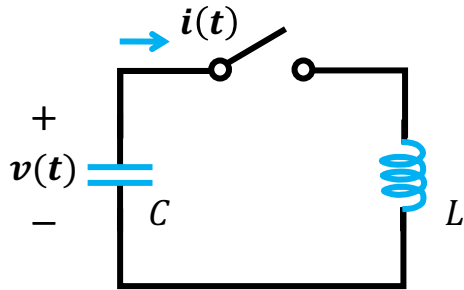
Steady state response (forced response) + Transient response (natural response) = Complete Response

Outlines

- 1st order circuit
 - Source free RC circuit
 - Source free RL circuit
 - Pulse response
- 2nd order circuits

Example 6: Source free LC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change after the switch is turned on.



- According to KCL

$$-C \frac{dv(t)}{dt} = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau$$

$$\rightarrow \frac{d^2 v(t)}{dt^2} + \frac{1}{LC} v(t) = 0$$

Recall: Calculus

QUESTION: Find the solution $x(t)$ of the following equation

$$\frac{d^2}{dt^2}x(t) + a_1 \frac{d}{dt}x(t) + a_2 x(t) = f(t) \quad \leftarrow \text{2nd order differential equation}$$

- Step 1: Find a solution $x(t) = x_c(t)$ to the homogeneous equation

$$\frac{d^2}{dt^2}x(t) + a_1 \frac{d}{dt}x(t) + a_2 x(t) = 0$$

Assume $x_c(t) = Ke^{st}$. Substitute it to the equation

$$s^2 Ke^{st} + a_1 s Ke^{st} + a_2 Ke^{st} = 0$$

Since $Ke^{st} \neq 0$

$$\begin{aligned} \rightarrow \quad s^2 + a_1 s + a_2 &= 0 \\ \text{Define } 2\zeta\omega_0 &= a_1, \omega_0^2 = a_2 \end{aligned} \quad \rightarrow \quad \begin{cases} S_1 = -\zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1} \\ S_2 = -\zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1} \end{cases}$$

Recall: Calculus

QUESTION: Find the solution $x(t)$ of the following equation

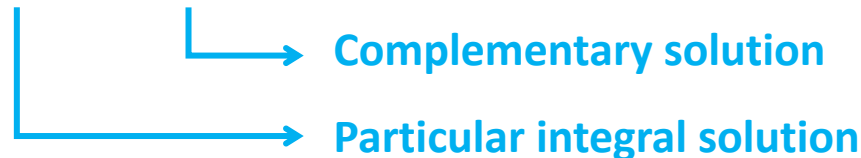
$$\frac{d^2}{dt^2}x(t) + a_1 \frac{d}{dt}x(t) + a_2 x(t) = f(t) \quad \text{Define } \begin{cases} 2\zeta\omega_0 = a_1 \\ \omega_0^2 = a_2 \end{cases}$$

- Step 1: Find a solution $x(t) = x_c(t)$ to the homogeneous equation

$$x_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t} \quad \text{where } \begin{cases} s_1 = -\zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1} \\ s_2 = -\zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1} \end{cases}$$

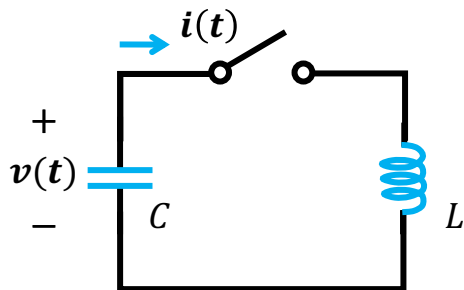
- Step 2: Find any solution to the original equation $x(t) = x_p(t)$
- Step 3: the solution to the original equation can be written as

$$x(t) = x_p(t) + x_c(t)$$



Example 6: Source free LC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change after the switch is turned on.



$$\frac{d^2 v(t)}{dt^2} + \frac{1}{LC} v(t) = 0$$

- Step 1a: Find the **particular integral solution** $v_p(t)$

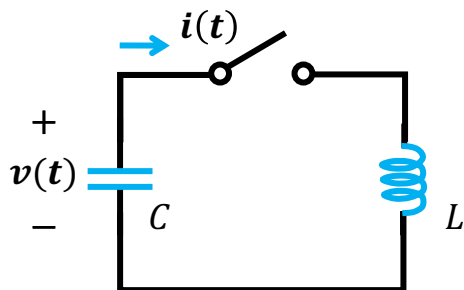
$$\text{Assume } v_p(t) = A$$

- Step 1b: substitute $v_p(t)$ to the equation

$$\frac{1}{LC} A = 0 \quad \rightarrow \quad A = 0$$

Example 6: Source free LC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change after the switch is turned on.



$$\frac{d^2 v(t)}{dt^2} + \frac{1}{LC} v(t) = 0$$

- Step 2a: find the **homogeneous equation**

$$\frac{d^2 v(t)}{dt^2} + \frac{1}{LC} v(t) = 0$$

- Step 2b: find the **complementary solution** $v_c(t)$ to the homogeneous equation

$$v_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

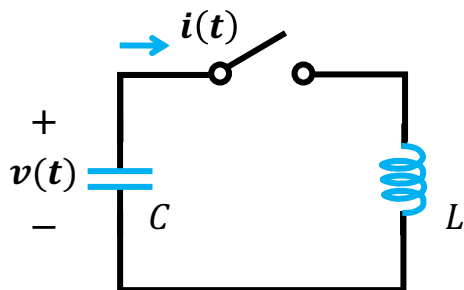
$$\begin{cases} s_1 = j\omega_0 \\ s_2 = -j\omega_0 \end{cases} \quad \text{where} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Particular integral solution

$$v_p(t) = 0$$

Example 6: Source free LC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change after the switch is turned on.



$$\frac{d^2 v(t)}{dt^2} + \frac{1}{LC} v(t) = 0$$

- Step 3a: find the value of $v(0)$ and $\left. \frac{dv(t)}{dt} \right|_{t=0}$
- Step 3b: substitute $v(0)$ and $\left. \frac{dv(t)}{dt} \right|_{t=0}$ to $v(t)$

Particular integral solution

$$v_p(t) = 0$$

Complementary solution

$$v_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

where $S_1 = j\omega_0$, $S_2 = -j\omega_0$

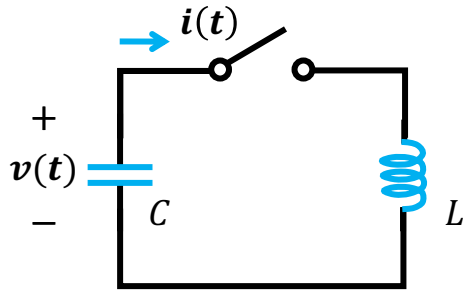
Full solution $v(t) = v_p(t) + v_c(t)$

$$\begin{cases} v(0) = K_1 + K_2 \\ \left. \frac{dv(t)}{dt} \right|_{t=0} = K_1 s_1 + K_2 s_2 = -\frac{i(0)}{C} \end{cases}$$

K_1 and K_2 can be solved

Example 6: Source free LC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change after the switch is turned on.

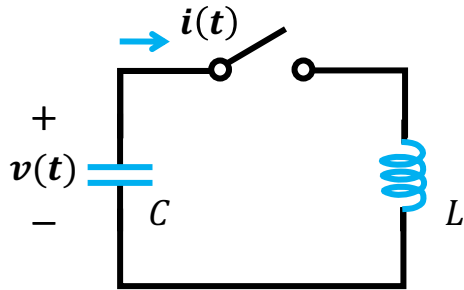


$$\begin{cases} v(t) = A \cos(\omega_0 t + \theta) \\ i(t) = \sqrt{\frac{C}{L}} A \sin(\omega_0 t + \theta) \end{cases}$$

$$\text{where } \begin{cases} A = \sqrt{v^2(0) + \frac{L}{C} i^2(0)} \\ \omega_0 = \frac{1}{\sqrt{LC}} \\ \theta = \tan^{-1} \left(\sqrt{\frac{L}{C}} \frac{i(0)}{v(0)} \right) \end{cases}$$

Example 6: Source free LC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change after the switch is turned on.



Energy stored in the capacitor/inductor

$$\begin{cases} w_C(t) = \frac{1}{2} C v^2(t) = \left(\frac{1}{2} C v^2(0) + \frac{1}{2} L i^2(t) \right) \cos^2(\omega_0 t + \theta) \\ w_L(t) = \frac{1}{2} L i^2(t) = \left(\frac{1}{2} C v^2(0) + \frac{1}{2} L i^2(t) \right) \sin^2(\omega_0 t + \theta) \end{cases}$$

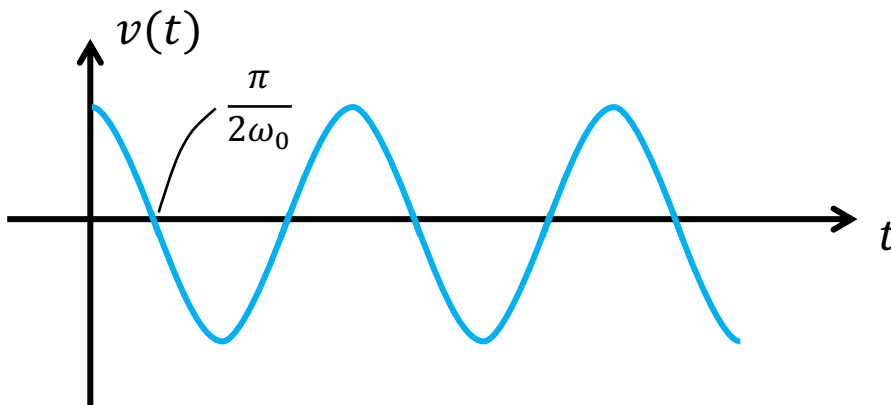
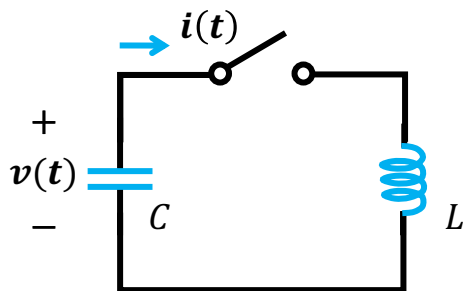
$$\begin{cases} v(t) = A \cos(\omega_0 t + \theta) \\ i(t) = \sqrt{\frac{C}{L}} A \sin(\omega_0 t + \theta) \end{cases}$$

The total energy doesn't change

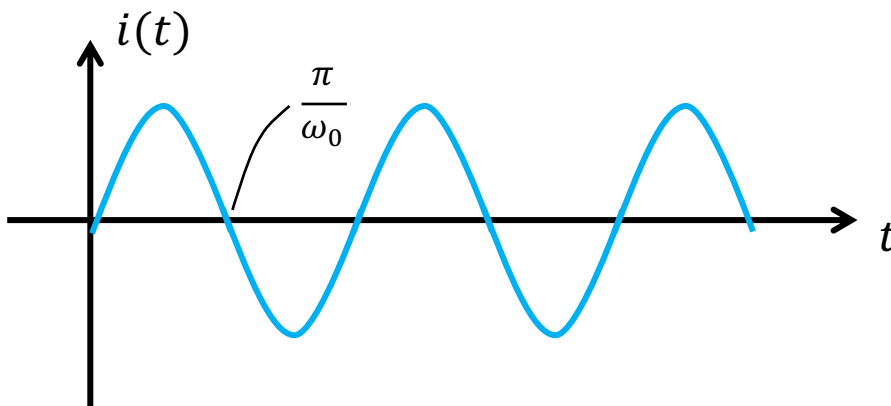
$$w(t) = w_C(t) + w_L(t) = \frac{1}{2} C v^2(0) + \frac{1}{2} L i^2(t)$$

Example 6: Source free LC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change after the switch is turned on.



$$\begin{cases} v(t) = A \cos(\omega_0 t + \theta) \\ i(t) = \sqrt{\frac{C}{L}} A \sin(\omega_0 t + \theta) \end{cases}$$

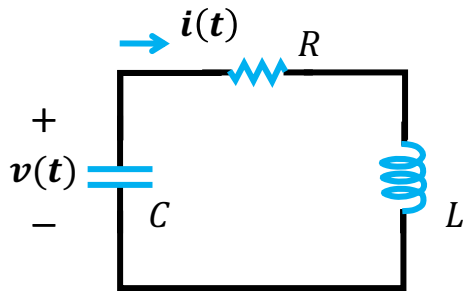


Outlines

- 1st order circuit
 - Source free RC circuit
 - Source free RL circuit
 - Pulse response
- 2nd order circuits
 - Source free LC circuit
 - **Source free RLC circuit**

Example 7: source free RLC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change.



- According to KCL

$$i(t) = -C \frac{dv(t)}{dt}$$

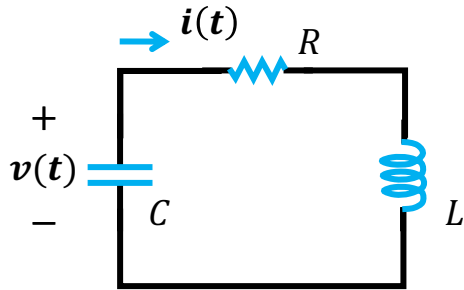
- According to KVL

$$-v(t) + i(t)R + L \frac{di(t)}{dt} = 0$$

$$\rightarrow \frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

Example 7: source free RLC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change.



$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

- Step 1a: Find the **particular integral solution** $v_p(t)$

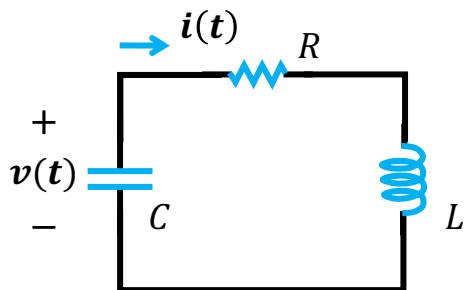
$$\text{Assume } v_p(t) = A$$

- Step 1b: substitute $v_p(t)$ to the equation

$$\frac{1}{LC} A = 0 \quad \rightarrow \quad A = 0$$

Example 7: source free RLC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change.



$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

- Step 2a: find the **homogeneous equation**

$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

- Step 2b: find the **complementary solution** $v_c(t)$ to the homogeneous equation

$$v_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

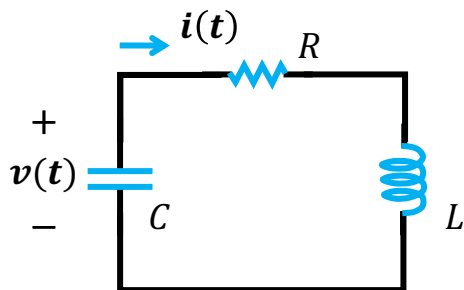
$$\rightarrow \begin{cases} s_1 = -\zeta \omega_0 + \omega_0 \sqrt{\zeta^2 - 1} \\ s_2 = -\zeta \omega_0 - \omega_0 \sqrt{\zeta^2 - 1} \end{cases} \quad \text{where} \quad \begin{cases} 2\zeta \omega_0 = \frac{R}{L} \\ \omega_0^2 = \frac{1}{LC} \end{cases}$$

Particular integral solution

$$v_p(t) = 0$$

Example 7: source free RLC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change.



$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

- Step 3a: find the value of $v(0)$ and $\left. \frac{dv(t)}{dt} \right|_{t=0}$
- Step 3b: substitute $v(0)$ and $\left. \frac{dv(t)}{dt} \right|_{t=0}$ to $v(t)$

Particular integral solution

$$v_p(t) = 0$$

Complementary solution

$$v_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

$$\text{Where } \begin{cases} s_1 = -\zeta \omega_0 + \omega_0 \sqrt{\zeta^2 - 1} \\ s_2 = -\zeta \omega_0 - \omega_0 \sqrt{\zeta^2 - 1} \end{cases}$$

$$\text{and } \omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$$

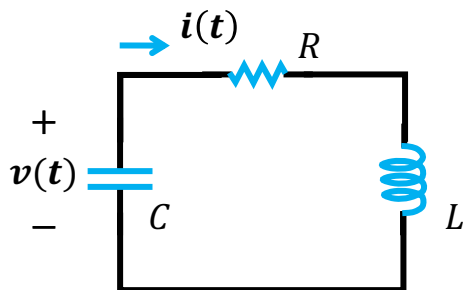
$$\text{Full solution } v(t) = v_p(t) + v_c(t)$$

$$\begin{cases} v(0) = K_1 + K_2 \\ \left. \frac{dv(t)}{dt} \right|_{t=0} = K_1 s_1 + K_2 s_2 = -\frac{i(0)}{C} \end{cases}$$

K_1 and K_2 can be solved

Example 7: source free RLC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change.



A DISCUSSION ON ζ

- If $\zeta = 1 \rightarrow$ **critically damped**

$$\rightarrow S_1 = S_2 = -\zeta\omega_0$$

$$v_c(t) = K_1 e^{-\zeta\omega_0 t} + K_2 t e^{-\zeta\omega_0 t}$$

- If $\zeta > 1 \rightarrow$ **overdamped**

$$v_c(t) = K_1 e^{(-\zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1})t} + K_2 e^{(-\zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1})t}$$

The response is the sum of two decaying exponentials

Particular integral solution

$$v_p(t) = 0$$

Complementary solution

$$v_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

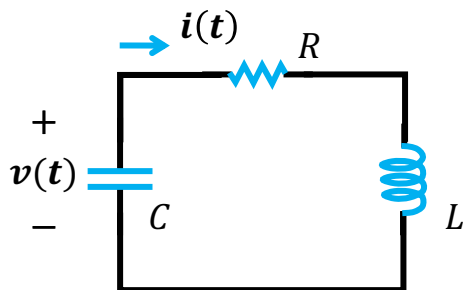
$$\text{Where } \begin{cases} S_1 = -\zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1} \\ S_2 = -\zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1} \end{cases}$$

$$\text{and } \omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$$

$$\text{Full solution } v(t) = v_p(t) + v_c(t)$$

Example 7: source free RLC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change.



A DISCUSSION ON ζ

- If $\zeta < 1 \rightarrow$ **underdamped**

$$\begin{cases} S_1 = -\zeta\omega_0 + j\omega_0\sqrt{1-\zeta^2} \\ S_2 = -\zeta\omega_0 - j\omega_0\sqrt{1-\zeta^2} \end{cases}$$

Define $\omega_d = \omega_0\sqrt{1-\zeta^2}$

$$v_c(t) = e^{-\zeta\omega_0 t} [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)]$$

The response is an exponentially damped oscillatory response

Particular integral solution

$$v_p(t) = 0$$

Complementary solution

$$v_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

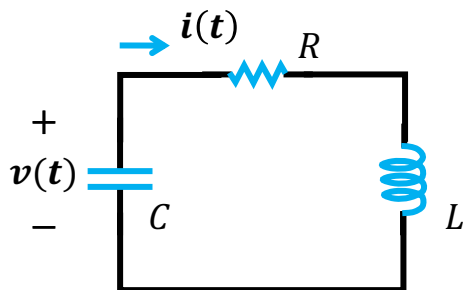
Where
$$\begin{cases} S_1 = -\zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1} \\ S_2 = -\zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1} \end{cases}$$

and $\omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$

Full solution $v(t) = v_p(t) + v_c(t)$

Example 7: source free RLC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change.



A DISCUSSION ON ζ

- If $\zeta < 1 \rightarrow$ **underdamped**
- If $\zeta = 1 \rightarrow$ **critically damped**
- If $\zeta > 1 \rightarrow$ **overdamped**

Particular integral solution

$$v_p(t) = 0$$

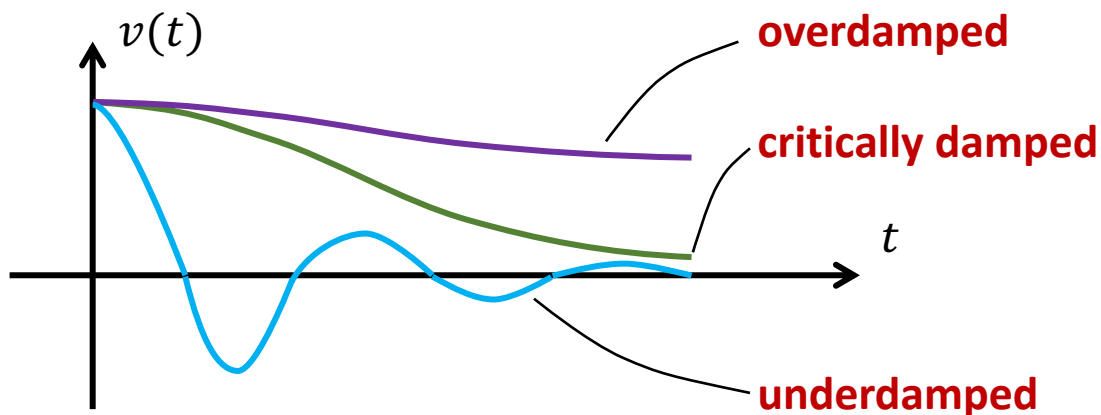
Complementary solution

$$v_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

Where
$$\begin{cases} s_1 = -\zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1} \\ s_2 = -\zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1} \end{cases}$$

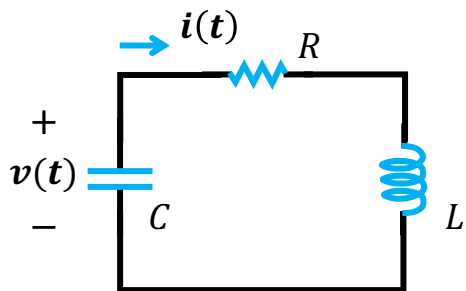
and $\omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$

Full solution $v(t) = v_p(t) + v_c(t)$



Example 7: source free RLC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change.



$$v_c(t) = e^{-\zeta\omega_0 t} [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)]$$

Define **DAMPING FACTOR**

$$\alpha = \frac{R}{2L}$$

Define **RESONATE FREQUENCY**

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Define **DAMPING RATIO**

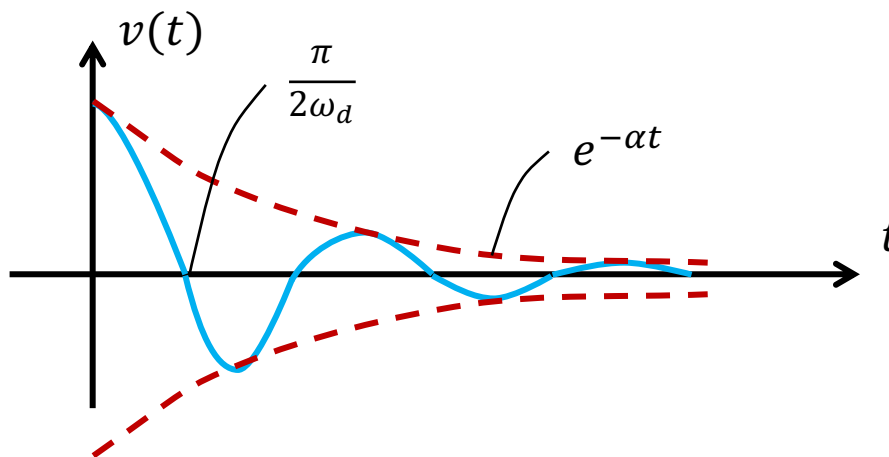
$$\zeta = \frac{\alpha}{\omega_0} = \frac{R}{2\omega_0 L}$$

$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

Solution: $v(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$

Where
$$\begin{cases} s_1 = -\zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1} \\ s_2 = -\zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1} \end{cases}$$

and $\omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$

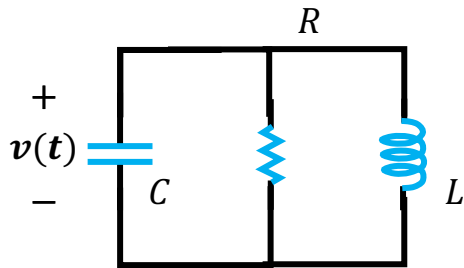


Outlines

- 1st order circuit
 - Source free RC circuit
 - Source free RL circuit
 - Pulse response
- 2nd order circuits
 - Source free LC circuit
 - Source free series RLC circuit
 - **Source free series parallel circuit**

Example 8: parallel connected RLC circuit

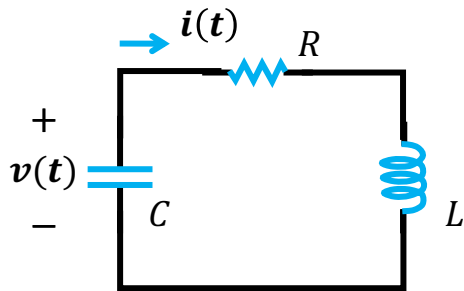
QUESTION: Find how the voltage on the capacitor and the current through the inductor change.



- According to KCL

$$C \frac{dv(t)}{dt} + \frac{v(t)}{R} + \frac{1}{L} \int_{-\infty}^t v(t) dt = 0$$

$$\rightarrow \frac{d^2 v(t)}{dt^2} + \frac{1}{RC} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

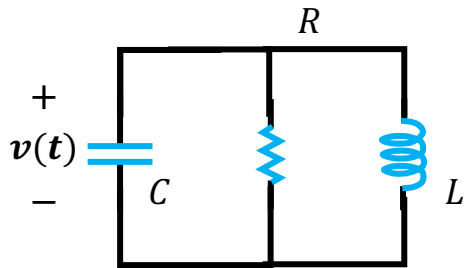


- Series connected RLC circuit

$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

Example 8: parallel connected RLC circuit

QUESTION: Find how the voltage on the capacitor and the current through the inductor change.

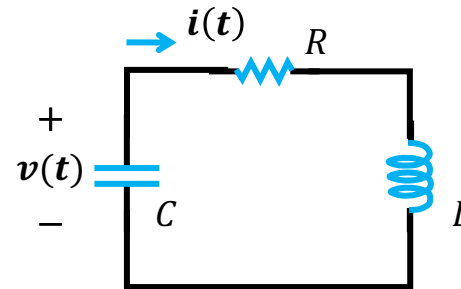


$$\frac{d^2 v(t)}{dt^2} + \frac{1}{RC} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

$$\text{Solution: } v(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

$$\text{Where } \begin{cases} s_1 = -\zeta \omega_0 + \omega_0 \sqrt{\zeta^2 - 1} \\ s_2 = -\zeta \omega_0 - \omega_0 \sqrt{\zeta^2 - 1} \end{cases}$$

$$\text{and } \omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{1}{2\omega_0 RC}$$



$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

$$\text{Solution: } v(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

$$\text{Where } \begin{cases} s_1 = -\zeta \omega_0 + \omega_0 \sqrt{\zeta^2 - 1} \\ s_2 = -\zeta \omega_0 - \omega_0 \sqrt{\zeta^2 - 1} \end{cases}$$

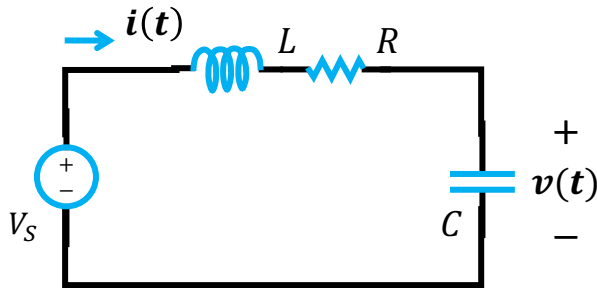
$$\text{and } \omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$$

Outlines

- 1st order circuit
 - Source free RC circuit
 - Source free RL circuit
 - Pulse response
- 2nd order circuits
 - Source free LC circuit
 - Source free series RLC circuit
 - Source free series parallel circuit
 - Response of RLC circuit

Example 9: RLC circuit w/ forcing

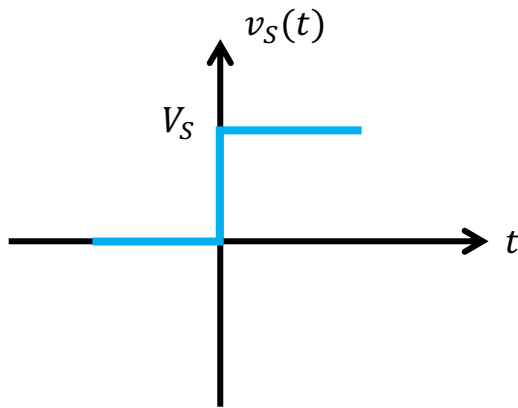
QUESTION: Find how the voltage on the capacitor and the current through the inductor change with an input voltage of step function.



- According to KVL, at $t > 0$

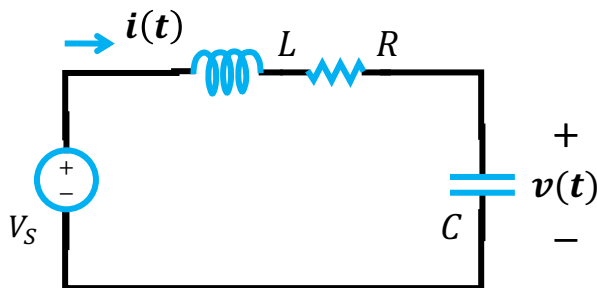
$$-v(t) + i(t)R + L \frac{di(t)}{dt} = V_s$$

$$\rightarrow \frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = \frac{V_s}{LC}$$



Example 9: RLC circuit w/ forcing

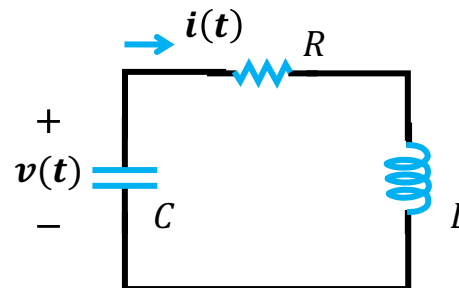
QUESTION: Find how the voltage on the capacitor and the current through the inductor change with an input voltage of step function.



- According to KVL

$$v(t) + i(t)R + L \frac{di(t)}{dt} = V_S$$

$$\rightarrow \frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = \frac{V_S}{LC}$$



$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

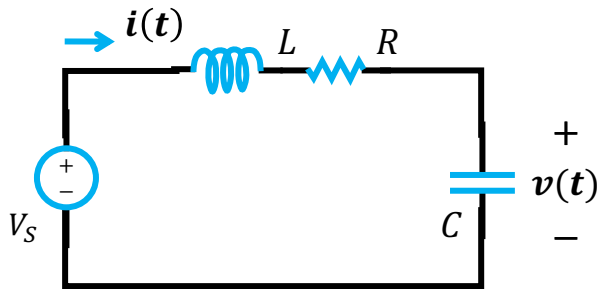
$$\text{Solution: } v(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

$$\text{Where } \begin{cases} s_1 = -\zeta \omega_0 + \omega_0 \sqrt{\zeta^2 - 1} \\ s_2 = -\zeta \omega_0 - \omega_0 \sqrt{\zeta^2 - 1} \end{cases}$$

$$\text{and } \omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$$

Example 9: RLC circuit w/ forcing

QUESTION: Find how the voltage on the capacitor and the current through the inductor change with an input voltage of step function.



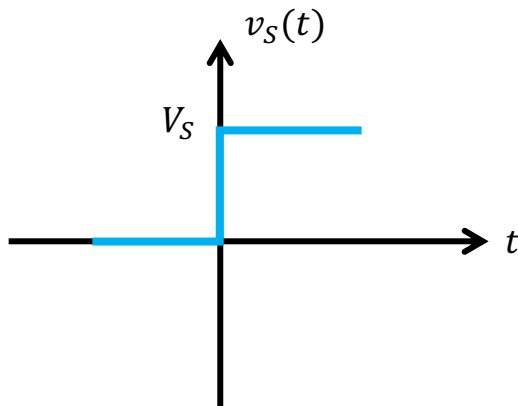
$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = \frac{V_S}{LC}$$

- Step 1a: Find the **particular integral solution** $v_p(t)$

$$\text{Assume } v_p(t) = A$$

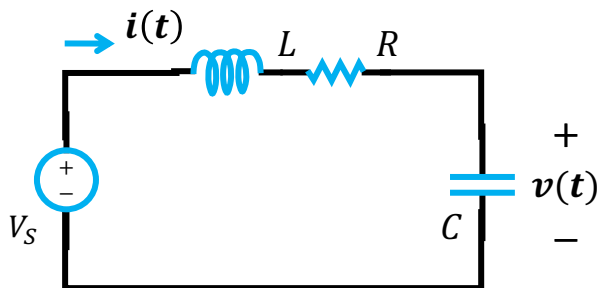
- Step 1b: substitute $v_p(t)$ to the equation

$$\frac{1}{LC} A = \frac{V_S}{LC} \quad \rightarrow \quad A = V_S$$



Example 9: RLC circuit w/ forcing

QUESTION: Find how the voltage on the capacitor and the current through the inductor change with an input voltage of step function.



Particular integral solution

$$v_p(t) = V_S$$

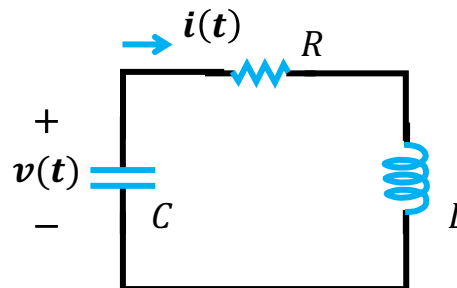
Complementary solution

$$v_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

$$\text{Where } \begin{cases} s_1 = -\zeta \omega_0 + \omega_0 \sqrt{\zeta^2 - 1} \\ s_2 = -\zeta \omega_0 - \omega_0 \sqrt{\zeta^2 - 1} \end{cases}$$

$$\text{and } \omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$$

$$\text{Full solution } v(t) = v_p(t) + v_c(t)$$



Particular integral solution

$$v_p(t) = 0$$

Complementary solution

$$v_c(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$$

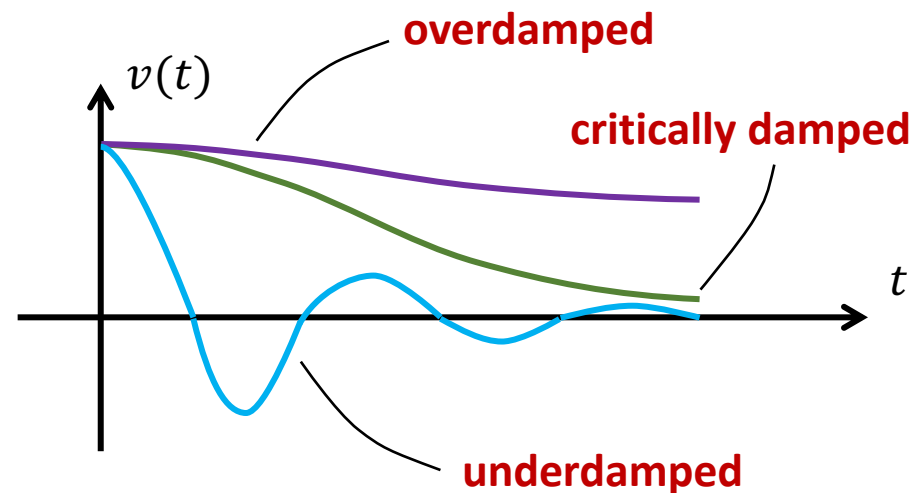
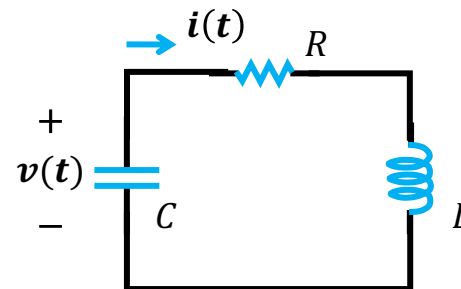
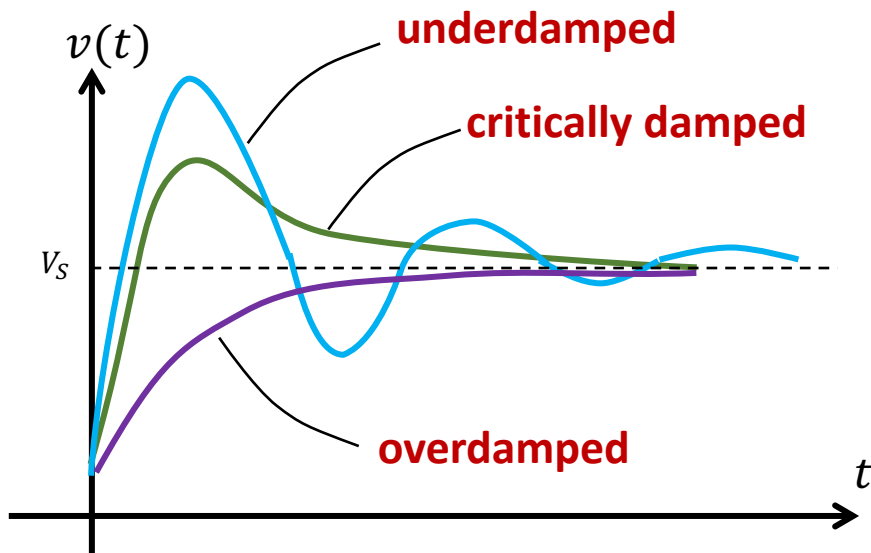
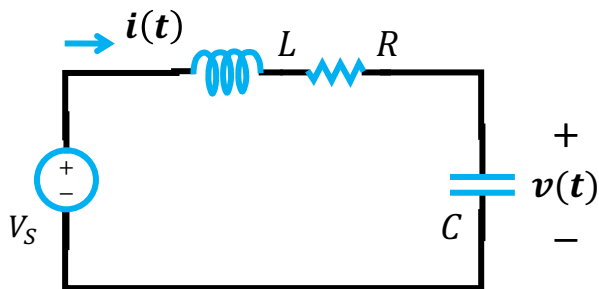
$$\text{Where } \begin{cases} s_1 = -\zeta \omega_0 + \omega_0 \sqrt{\zeta^2 - 1} \\ s_2 = -\zeta \omega_0 - \omega_0 \sqrt{\zeta^2 - 1} \end{cases}$$

$$\text{and } \omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$$

$$\text{Full solution } v(t) = v_p(t) + v_c(t)$$

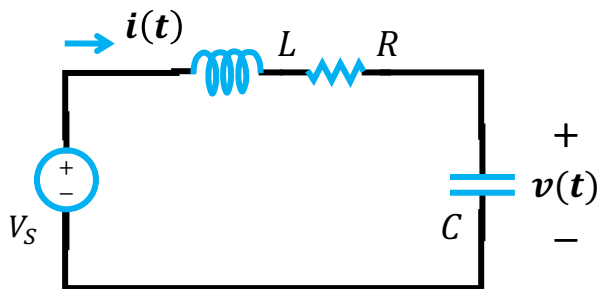
Example 9: RLC circuit w/ forcing

QUESTION: Find how the voltage on the capacitor and the current through the inductor change with an input voltage of step function.



Example 9: RLC circuit w/ forcing

QUESTION: Find how the voltage on the capacitor and the current through the inductor change with an input voltage of step function.



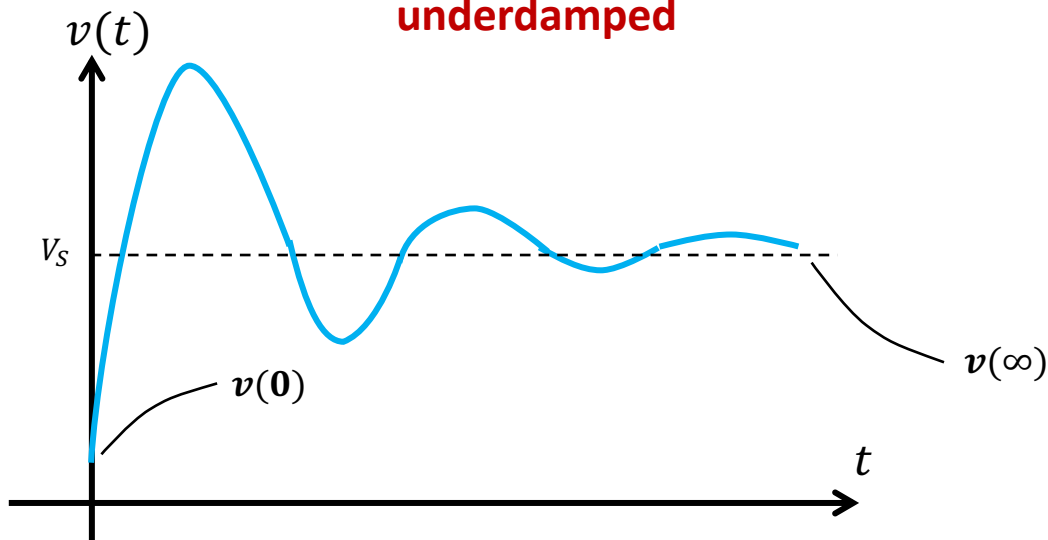
$$v(t) = \underbrace{V_S}_{\text{Steady state response (forced response)}} + \underbrace{e^{-\zeta\omega_0 t} [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)]}_{\text{Transient response (natural response)}}$$

Steady state response
(forced response)

+ Transient response
(natural response)

= Complete Response

underdamped



Outlines

- How to analyze 1st/2nd order circuit in time domain?
 - Write the circuit equation according to KVL/KCL
 - Solve the differential equation
 - Step 1a: Find the **particular integral solution** $v_p(t)$
 - Step 1b: substitute $v_p(t)$ to the equation to solve the unknown
 - Step 2a: find the **homogeneous equation**
 - Step 2b: find the **complementary solution** $v_c(t)$ to the homogeneous equation
 - Step 3a: find the initial voltage/current values
 - Step 3b: substitute the initials to the full solution to solve the unknown

Reading tasks & learning goals

- Reading tasks

- Basic Engineering Circuit Analysis, 10th edition
 - Chapter 7

- Learning goals

- Be able to calculate initial values for inductor currents & capacitor voltages in transient circuits
- Be able to calculate V/I in 1st order transient circuit
- Be able to calculate V/I in 2nd order transient circuit
- Know what is time constant, steady state response, transient response, complete response